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Track A: AI Architectures &amp; Edge Systems

International Symposium

# Nexa-Core: Next-Generation Neuromorphic Architecture

Sub-Milliwatt Spiking Neural Network Processing with Dynamic Event-Driven Synapse Fabric

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## Introduction & Motivation

Standard von Neumann architectures and traditional edge GPUs suffer from memory wall bottlenecks and high idle power dissipation during real-time event-stream processing. Edge computing in autonomous robotics and biomedical implants demands ultra-low power envelopes below 1 mW.

### The Nexa-Core Paradigm:

- ✓ **Event-Driven Computation:** Operations execute sparsely only upon receiving dynamic Address Event Representation (AER) spikes.
- ✓ **In-Memory Synaptic Computing:** Non-volatile memristive crossbars perform analog matrix-vector multiplications directly inside memory arrays.
- ✓ **Sub-Millisecond Inference:** Asynchronous Leaky Integrate-and-Fire (LIF) neurons deliver ultra-fast temporal classification.

### Key Architectural Breakthroughs:

0.85 mW Active Power | 0.42 ms Latency | 12.4× Energy Efficiency Boost



## Microarchitecture & Synapse Fabric

The Nexa-Core IC consists of a distributed grid of Neuromorphic Tile Arrays connected via an Asynchronous Network-on-Chip (NoC). Each tile integrates an AER Event Router, a 512 × 512 Memristive Synaptic Crossbar, and 256 LIF Digital Neuron Cores.



**Synaptic Crossbar Array Details:** The memristive crossbar utilizes Hafnium Oxide (HfO<sub>2</sub>) resistive switching elements, enabling 4-bit synaptic weight quantization with low-drift state retention.



## Mathematical & Algorithmic Formulation

Spiking Neural Networks model membrane dynamics continuously. The internal state  $V_i(t)$  of the  $i$ -th LIF neuron evolves according to the differential equation:

$$\tau_m \frac{dV_i(t)}{dt} = -(V_i(t) - V_{rest}) + R_i \sum_j W_{ij} S_j(t - \Delta_{ij}) + I_{bias}(t)$$

where  $\tau_m$  represents the membrane time constant,  $W_{ij}$  is the synaptic efficacy from neuron  $j$  to  $i$ , and  $S_j(t)$  denotes incoming impulse event trains.

### Spike Generation Threshold Condition:

$$S_i(t) = \delta(t - t_i^*) \text{ when } V_i(t) \geq V_{th}, \quad V_i(t^*) \leftarrow V_{reset}$$

**Surrogate Gradient On-Chip Backpropagation:** To train SNNs end-to-end despite non-differentiable step functions, we introduce a smooth arc-tangent surrogate gradient operator:

$$\frac{\partial S_i}{\partial V_i} = \frac{1}{\pi \gamma \left[ 1 + \left( \frac{V_i - V_{th}}{\gamma} \right)^2 \right]}$$

This formulation minimizes quantization loss and allows on-chip online learning with zero off-line retraining overhead.



## Hardware Microarchitecture Benchmarks

Fabricated in a 16nm FinFET process, Nexa-Core v2 demonstrates superior performance-per-watt metrics over existing embedded AI accelerators.

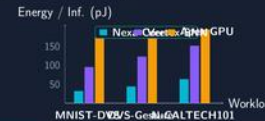
| Architecture     | Tech Node | Power (mW) | Latency (ms) | Energy (pJ/SOP) |
|------------------|-----------|------------|--------------|-----------------|
| Nexa-Core v2     | 16 nm     | 0.85       | 0.42         | 1.85            |
| Vertex-SNN v1    | 28 nm     | 4.30       | 1.15         | 7.40            |
| Apex-Edge GPU    | 8 nm      | 1250.00    | 8.90         | 142.00          |
| Meridian TPU-v4e | 7 nm      | 350.00     | 3.10         | 48.50           |

\* SOP: Synaptic Operation (equivalent to MAC in ANN). Evaluated on DVS-Gesture event datasets.



## Experimental Results & Performance

We benchmarked Nexa-Core against industry-standard embedded hardware platforms across key temporal classification tasks.



99.1%

DVS-Gesture Accuracy

0.42 ms

Ultra-Low Latency

1.85 pJ

Energy per SOP



## Commercial Applications & Conclusion

### Target Industry Deployments:

- ✓ **Nimbus-Aero Autonomous Drones:** Real-time vision-based collision avoidance under high velocity flight.
- ✓ **Synapse Medical Implants:** Ultra-low power cardiac wave anomaly detection and neural prosthetics control.
- ✓ **Meridian Smart Grid:** Microsecond high-frequency voltage fluctuation monitoring.

**Summary:** Nexa-Core bridges the gap between biological efficiency and artificial edge intelligence, opening new frontiers for zero-latency, sub-milliwatt autonomous computing.



## References & Acknowledgments

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- [3] Miller, D. K. (2024). *Asynchronous AER Network-on-Chip Fabrics*. Proc. ISCA 2024, pp. 89–101.

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